Syllabus Details

Syllabus ID:	11424
Syllabus Name:	**Introduction to Electronic Components and Circuits_Nhập môn các linh kiện và mạch điện tử**
Course Name English:	**Introduction to Electronic Components and Circuits**
Subject Code:	**ECI101**
Learning-Teaching Method:	Inquiry-Based Learning, In-class lecture, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 90h self-study
Pre-Requisite:	
Description:	This course has been designed to help students understand how electronic circuits work. It will provide them with the basic underpinning knowledge necessary to appreciate the operation of a wide range of electronic circuits, including amplifiers, logic circuits, power supplies and oscillators. The main focus of this course is not only to develop the student ability to analyze and design basic analog electronic circuits with passive components and/or the active elements like diode and transistors. But also, some experiments are provided to help students to have a thorough grasp of the basic electronic circuit problems. During the course, Multisim, the supporting software for circuit analysis is also illustrated appropriately.
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all exercises and practical given by instructor in class or at home and submit on time. Do quizzes during class. - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	Specific Tools: 1) Simulation software - MultisimLive: https://www.multisim.com (online simulation) - Tinkercad.com - Tina Circuits Simulator 2) Lab equipments/ Training Kits - Tool: Oscilloscope and Function generator, multiMenter - Kit for lab: List o
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	358/QĐ-ĐHFPT dated 04/03/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/3/2024

3 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Electronic Circuits Fundamentals and Applications	Mike Tooley	Routledge	2006	3 rd	ISBN-13: 978-0-75-066923-8	True	False	True	Link: https://library.books24×7.com/toc.aspx?bookid=32309

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
2	Electronic Circuits Fundamentals and Applications	Mike Tooley	Routledge	2019	5 th	ISBN: 978-0-367-42199-1	False	True	False	
3	Electronic Devices and Circuit Theory	Robert L. Boylestad, Louis Nashelsky	Pearson	2013	11 th	ISBN: 9780132622264	False	True	False	

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Explain the theoretical principles essential for understanding the operation of electronic circuits
2	CLO2	Understand passive and active electronic components and their applications
3	CLO3	Analyze electrical circuits and calculate the main parameters
4	CLO4	Explain the technological principles of semiconductor manufacturing
5	CLO5	Develop, design and create simple analogue and digital electronic circuits
6	CLO6	Measure the characteristics of electronic circuits and present experimental results
7	CLO7	Choose an engineering approach to solving problems, starting from the acquired knowledge essential for the design of electronic circuits
8	CLO8	Demonstrate professional ethics, social responsibility, and environmental impacts into electronic component selection, design, and use throughout the entire product life cycle

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course overview Chapter 1: Electrical fundamentals 1.1. Fundamentals units (SI)	Offline	CLO1	TU	Syllabus, textbook	ECI101	Read & Review Slides (Chapter 1 - 1.1)	
2	1.2. Conductors, insulators, voltage and resistance 1.3. Ohm's Law 1.4. Electric fields 1.5. Magnetic fields 1.6. Circuit diagrams	Offline	CLO1	TU	Syllabus, textbook		Read & Review Slides (Chapter 1 - 1.2, 1.3, 1.4, 1.5, 1.6)	
3	1.7. Practical investigation & Problems	Offline	CLO1	TU	Syllabus, textbook		Read & Review Slides (Chapter 1 - 1.7)	
4	Chapter 2: Passive components 2.1. Resistors 2.2. Capacitance	Offline	CLO2	TU	Syllabus, textbook		Read & Review Slides (Chapter 2 - 2.1, 2.2)	
5	2.3. Inductors	Offline	CLO2	TU	Syllabus, textbook		Read & Review Slides (Chapter 2 - 2.3)	
6	2.4. Practical investigation & Problems	Offline	CLO2	TU	Syllabus, textbook		Read & Review Slides (Chapter 2 - 2.4)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
7	Chapter 3: D.C. circuits 3.1. The Theorems and Laws	Offline	CLO3	TU	Syllabus, textbook		Read & Review Slides (Chapter 3 - 3.1)	
8	3.2. L-R-C circuits	Offline	CLO3	TU	Syllabus, textbook		Read & Review Slides (Chapter 3 - 3.2)	
9	3.3. Practical investigation & Problems	Offline	CLO3	TU	Syllabus, textbook		Read & Review Slides (Chapter 3 - 3.3)	
10	Chapter 4: Alternating voltage and current 4.1. Alternating versus direct current 4.2. Reactance	Offline	CLO3	TU	Syllabus, textbook		Read & Review Slides (Chapter 4 - 4.1, 4.2)	
11	4.3. Power factor and Resonance 4.4. Transformers	Offline	CLO3	TU	Syllabus, textbook		Read & Review Slides (Chapter 4 - 4.3, 4.4)	
12	4.5. Practical investigation & Problems	Offline	CLO3	TU	Syllabus, textbook		Read & Review Slides (Chapter 4 - 4.5)	
13	Chapter 5: Semiconductors 5.1. Semiconductor junction and diodes 5.2. Bipolar junction transistors	Offline	CLO2, CLO4	TU	Syllabus, textbook		Read & Review Slides (Chapter 5 - 5.1, 5.2)	
14	5.3. Field effect transistors 5.4. Integrated circuits	Offline	CLO2, CLO4	TU	Syllabus, textbook		Read & Review Slides (Chapter 5 - 5.3, 5.4)	
15	5.5. Practical investigation & Problems	Offline	CLO2, CLO4	TU	Syllabus, textbook		Read & Review Slides (Chapter 5 - 5.5)	
16	Chapter 6: Power supplies 6.1. Rectifiers	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 6 - 6.1)	
17	6.2. Practical power supply circuits	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 6 - 6.2)	
18	6.3. Practical investigation & Problems	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 6 - 6.3)	
19	Lab 1	Offline	All CLOs	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
20	Lab 1	Offline	All CLOs	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
21	Lab 1	Offline	All CLOs	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
22	Chapter 7: Amplifiers 7.1. Transistor amplifiers	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 7 - 7.1)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
23	7.2. Equivalent circuits 7.3. Equivalent circuits	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 7 - 7.2, 7.3)	
24	7.4. Practical investigation & Problems	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 7 - 7.4)	
25	Chapter 8: Operational amplifiers 8.1. Operational amplifier parameters, characteristics and applications	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 8 - 8.1)	
26	8.2. Operational amplifier circuits	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 8 - 8.2)	
27	8.3. Practical investigation & Problems	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 8 - 6.3)	
28	Exercise 1	Offline	CLO1-5, CLO8	TU	Syllabus, textbook		Complete the exercises	
29	Chapter 9: Oscillators 9.1. Positive feedback and conditions for oscillation 9.2. Some Typical Oscillator Circuits	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 9 - 9.1, 9.2)	
30	9.3. The astable multivibrator 9.4. Practical oscillator circuits & Problems	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 9 - 9.3, 9.4)	
31	Chapter 10: Logic circuits 10.1. Basic logic functions and gates 10.2. Types Bistables	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 10 - 10.1, 10.2)	
32	10.3. Integrated circuit logic devices	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 10 - 10.3)	
33	10.4. Practical investigation & Problems	Offline	CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 10 - 10.4)	
34	Progress test 1	Offline	CLO1-5	TU	Syllabus, textbook		Complete the quiz	
35	Chapter 12: The 555 timer 12.1. The 555 family 12.2. Monostable pulse generator	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 12 - 12.1, 12.2)	
36	12.3. A variable pulse generator 12.4. Practical investigation & Problems	Offline	CLO2, CLO5	TU	Syllabus, textbook		Read & Review Slides (Chapter 12 - 12.3, 12.4)	
37	Chapter 14: Test equipment and measurements 14.1. Multi-range meters 14.2. Digital multi-range meters	Offline	CLO6	TU	Syllabus, textbook		Read & Review Slides (Chapter 14 - 14.1, 14.2)	
38	14.3. The oscilloscope	Offline	CLO6	TU	Syllabus, textbook		Read & Review Slides (Chapter 14 - 14.3)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
39	14.4. Practical investigation & Problems	Offline	CLO6	TU	Syllabus, textbook		Read & Review Slides (Chapter 14 - 14.4)	
40	Lab 2	Offline	All CLO	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
41	Lab 2	Offline	All CLO	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
42	Lab 2	Offline	All CLO	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
43	Chapter 15: Fault finding 15.1. Approach to fault finding Diode, Transistor and Integrated circuit	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 15 - 15.1)	
44	15.2. A transistor/diode checker 15.3. Logic circuit faults	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 15 - 15.2, 15.3)	
45	15.4. Practical investigation & Problems	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 15 - 15.4)	
46	Chapter 16: Sensors and interfacing 16.1. Instrumentation and control system	Offline	CLO2	TU	Syllabus, textbook		Read & Review Slides (Chapter 16 - 16.1)	
47	16.2. Transducers and Sensors 16.3. Outputs and consumption load	Offline	CLO2	TU	Syllabus, textbook		Read & Review Slides (Chapter 16 - 16.2, 16.3)	
48	16.4. Practical investigation & Problems	Offline	CLO2	TU	Syllabus, textbook		Read & Review Slides (Chapter 16 - 16.4)	
49	Chapter 17: Circuit simulation 17.1. Types of analysis 17.2. Netlists and component models	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 17 - 17.1, 17.2)	
50	17.3. Logic simulation	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 17 - 17.3)	
51	17.4. Practical investigation & Problems	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 17 - 17.4)	
52	Chapter 19: Circuit construction 19.1. Circuit construction methods 19.2. Solder and soldering technique	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 19 - 19.1, 19.2)	
53	19.3. Heatsinks 19.4. Practical investigation & Problems	Offline	CLO7	TU	Syllabus, textbook		Read & Review Slides (Chapter 19 - 19.3, 19.4)	
54	Exercise 2	Offline	CLO5-8	TU	Syllabus, textbook		Complete the exercises	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
55	Lab 3	Offline	All CLOs	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
56	Lab 3	Offline	All CLOs	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
57	Lab 3	Offline	All CLOs	TU	Simulation software and Lab equipments/ Training Kits		Complete the lab exercises	
58	Progress test 2	Offline	CLO5-7	TU	Syllabus, textbook		Complete the quiz	
59	Review	Offline	All CLOs	TU	Syllabus, textbook		Read & Review Slides (All chapters)	
60	Review	Offline	All CLO	TU	Syllabus, textbook		Read & Review Slides (All chapters)	

0 Constructive question(s)

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment / Lab	on-going	3	30.0%	>0	135'	All CLOs	Electronic Circuit Design, Analysis and Implementation	1 - 2	Lab1: Covering knowledge from Chapter 1 to Chapter 6 Lab2: Covering knowledge from Chapter 7 to Chapter 14 Lab3: Covering knowledge from Chapter 15 to Chapter 19	Individual or team work, guided by instructor, submission by a given deadline	
2	Exercises	on-going	2	20.0%	>0	40' - 60'	EX1: CLO1-5, CLO8; EX2: CLO1-65, CLO8; Number of questions: 1-2 for each CLOx	Writing	3 - 8	EX1: Covering knowledge from Chapter 1 to Chapter 8 EX2: Covering knowledge from Chapter 9 to Chapter 19	supervised by instructor	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
3	Progress test	on-going	2	20.0%	>0	30' - 40'	PT1: CLO1-5 PT2: CLO5-7 Number of questions: 3-8 for each CLOx	Multiple choice	25 - 40	PT1: Covering knowledge from Chapter 1 to Chapter 10 PT2: Covering knowledge from Chapter 11 to Chapter 19	supervised by instructor	
4	Final exam	Final exam	1	30.0%	4	60'	All CLOs	Multiple choice	50	All chapters; knowledge and skills	by Exam board	